Earth and Planetary Science Letters, 98 (1990) 329–339
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[XLeP]

Post-Oligocene rotations in southern Ecuador and northern Peru and the formation of the Huancabamba deflection in the Andean Cordillera

Pierre Mitouard, Catherine Kissel and Carlo Laj

Centre des Faibles Radioactivités, Laboratoire Mixte CNRS-CEA, Avenue de la Terrasse, 91198 Gif-sur-Yvette Cedex (France)

Received September 15, 1989; revised version accepted February 15, 1990

A paleomagnetic study of more than 410 samples from 41 sites from Paleocene to early Oligocene volcanic and intrusive formations in southern Ecuador and northern Peru reveals a pattern of rotations in opposite senses on both sides of the Huancabamba deflection. The amplitude of the rotations is ~ 25° clockwise north of the deflection and ~ 20° anticlockwise south of it. North of the deflection, the results do not allow us to ascertain whether the rotation arises from oroclinal bending or from block rotation in a distributed shear. South of it, the absence of geological evidence for widespread strike-slip faulting suggests that the results reflect a rotation of the Peruvian margin which could be related to the shortening documented in Southern Peru and to the uplift of the Central Andes.

1. Introduction

In the last decade paleomagnetic measurements have been widely used for studying the tectonic processes along active margins, such as the Mediterranean or the North American Cordilleras. On the contrary, in South America, only a limited amount of paleomagnetic studies has been done until recently, despite the rather large body of geological and geophysical work undertaken over the past twenty years or more to investigate the building and shaping of the Andean Cordillera.

Extensive paleomagnetic studies have indeed only been made in the last few years. Many of these studies concern the region of the Central Andes in southern Peru and northern Chile where the trend of the Cordillera undergoes a significant change known as the Arica elbow. Because the geological studies in this region have not recognized any ophiolitic suture, the Central Andes have generally been considered as a genuine marginal orogen, related exclusively to subduction. Accordingly, previous paleomagnetic results, which indicate counterclockwise rotations in Peru and northernmost Chile and clockwise rotation in Chile, have generally been interpreted in terms of bending of the Bolivian orocline [1–4]. Beck [5] has, however, recently proposed an alternative explanation in terms of in-situ block rotations in response to shear (sinistral to the north and dextral to the south of the Arica elbow). In this model, the coastal rotations are related to the geometry of the convergence between oceanic and continental plates.

In contrast, no constraining paleomagnetic results have yet been obtained about the origin and evolution of the other major bend of the Andean Cordillera, known as the Huancabamba deflection, situated farther north at 4° S (Fig. 1). In this region the trend of the Huancabamba Andes, which extend between 3° S and 8° S from southern Ecuador to northern Peru, changes from N020 in the north to N140 in the south.

The only available paleomagnetic results from the Huancabamba Andes, obtained by our group [6], concern Paleozoic and Cretaceous units from the Amotape–Tahuin massif and the Lancones basin north of the Huancabamba deflection. We interpreted these results in terms of the accretion of an allochthonous continental terrane at Neocomian time, followed by post-accretion in-situ rotation during the Cretaceous. The total clockwise rotation is ~ 90° since the Aptian and ~ 65° since the Senonian. Unexpectedly, a paleomagnetic direction indicating a ~ 35° clockwise rotation was obtained from a site sampled in an ill-dated but most probably Paleocene granodioritic pluton. No sound tectonic conclusion

0012-821X/90/$03.50 © 1990 – Elsevier Science Publishers B.V.

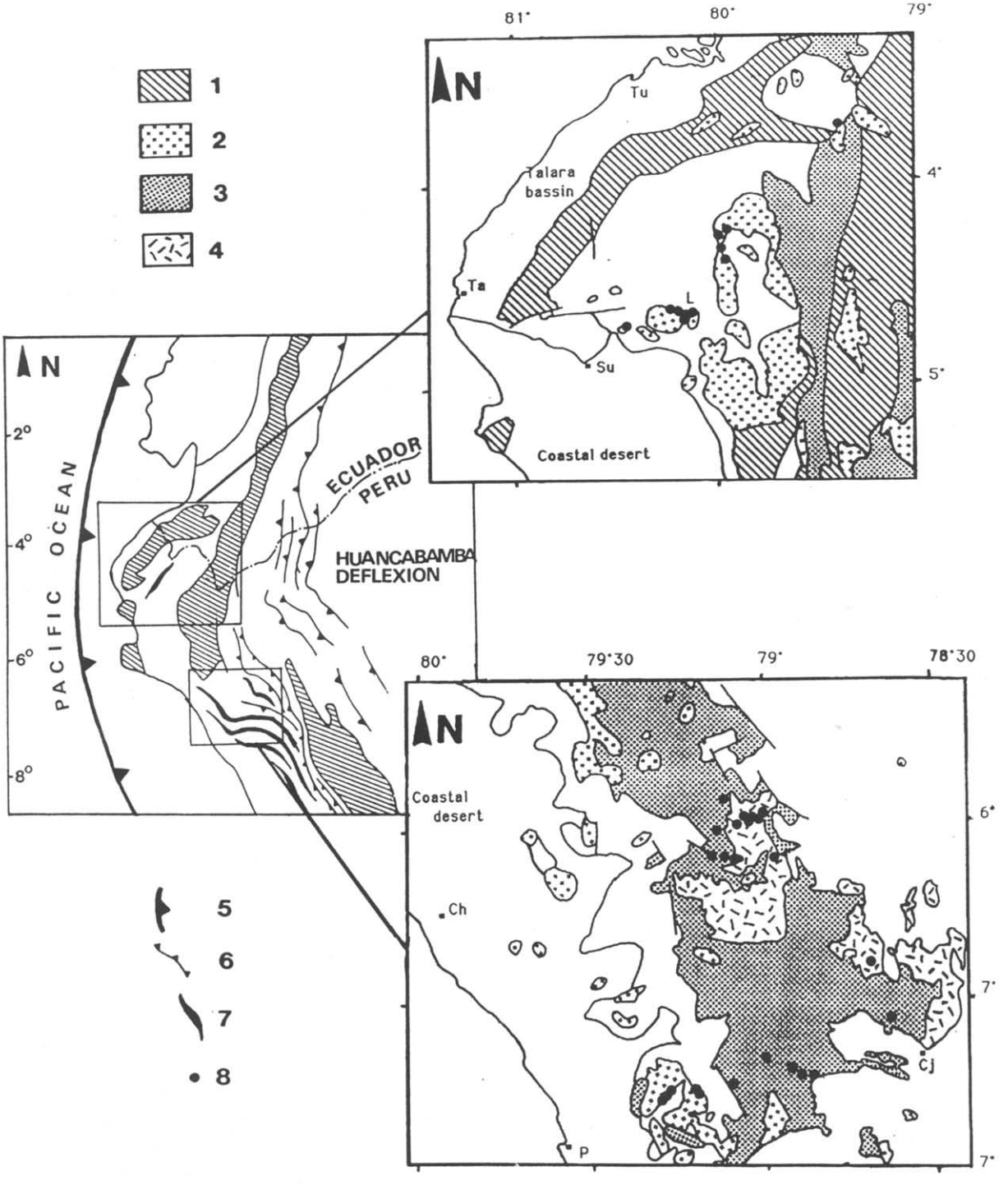


Fig. 1. Schematic map of the Huancabamba Andes showing the location of the paleomagnetic sites. *1* = Precambrian/Paleozoic basement; *2* = Post Cretaceous intrusives; *3* = Meso-Cenozoic undifferentiated volcanic formations; *4* = Meso-Cenozoic undifferentiated series; *5* = Huambos volcanic formation; *6* = Present trench; *7* = main thrusts; *8* = Axes of major folds; *9* = Sampling sites; *Ch* = Chiclayo; *Cj* = Cajamarca; *L* = Las Lomas; *P* = Pacasmayo; *Su* = Sullana; *Ta* = Talara; *Tu* = Tumbes.

could be obtained from this single site, but this result prompted additional sampling to further investigate the hypothesis of a post-Paleocene rotation of the Lancones Basin. Some preliminary results from these additional sites have already been reported [7]. Here we give a more complete report of the results obtained from all sites in the Lancones Basin, including a description of rock magnetic properties. We also discuss the results obtained from volcanic and intrusive formations of similar age situated farther south in the Cajamarca region, in order to compare the sense and the amount of post-Paleocene rotations, north and south of the Huancabamba deflection.

2. Geological setting and paleomagnetic sampling

The Lancones synclinorium is situated in northern Peru and southern Ecuador between the Amotape–Tahuin range and the Olmos massif, north of the Huancabamba deflection. In its western part, the Cretaceous formations consist of Albian carbonates overlain by a flysch series of late Cretaceous age. They grade eastward into mixed volcanic and partly volcaniclastic sedimentary rocks. These volcanics are characterized by pillow-lava flows intercalated with hyaloclastic breccias and scarce volcaniclastic strata, intruded by dykes and sills of basaltic to andesitic composition. Although not yet dated radiometrically, these pillow lavas are constrained in age to be pre-Albian. They are unconformably overlain by formations of the Albian to Senonian volcanic arc and by volcaniclastic series, both of which are intruded by granodioritic plutons of post-Senonian and pre-Oligocene age [8,9]. Although this plutonism is not precisely dated, two different units can be distinguished, the younger unit having a higher Si content [10].

The Cajamarca area, at about 7°S, is the southern limit of the Huancabamba deflection. There volcanic formations overlie the Mesozoic deformed formations of the western cordillera. These volcanic formations consist mainly of two units: the deformed Llama formation and the overlying almost undeformed Huambos formation. The main part of the Llama formation is constituted by thick andesitic lava flows which overlie volcanic sandstones and conglomerates. The thickness of this formation varies from 1500 to 2000 m. K/Ar ages estimated from these flows range between 54.8 and 44.2 Ma [11]. The Llama formation was deformed during the Incaïc tectonic phase (around 42 Ma) and then was covered by the Huambos formation dated by the K/Ar method to be between 35.4 and 38.4 Ma [11]. The basal unit of the latter is characterized by a pyroclastic flow of rhyolitic welded tuff and its upper unit by dacitic tuffs and volcanic sandstones. In the coastal zone of the Cajamarca area, the Mesozoic formations and the Cenozoic volcanic formations are intruded by a few granodioritic stocks which have been dated at 43 My [12].

We have sampled the intrusive formations of the Lancones basin and the intrusive units and the Huambos and Llama volcanics of the western Cajamarca area. Two sites were also sampled in an intermediate area near the village of Olmos (6°S) in the coastal intrusive formation. Over 410 cores were obtained from 41 sites whose location are shown in Fig. 1.

In the Lancones basin, five sites were sampled in Ecuador: one site near the village of Loja in a stock which intrudes andesitic lava from the Saraguro Group, and four sites in the Macara formation, which intrudes the Celica volcanic formation. In spite of careful search we have not succeeded in locating any fresh outcrops in the Celica formation in Ecuador. The other sites of this basin were sampled in northern Peru near Las Lomas: four sites in the Las Lomas Granodiorite, one site in the Montecillo Granite and one in the Penablanca Monzogranite. In the Cajamarca area, fifteen sites were sampled in the Llama formation and eight sites in the Huambos volcanics near Llama, Huambos and Sexi. Difficulty in gaining access to outcrops precluded more exhaustive sampling. Near Tembladera, five sites were obtained from the Gallito Ciego post-tectonic granodioritic intrusions dated at 43 My [12].

We have assumed that these post-tectonic intrusions have not been significantly tilted on the basis of both published and unpublished geological observations [10] (F. Mégard, pers. commun., 1987). Tectonic bedding attitudes could be measured in the Llama formation because, in most of the cases, volcano-sedimentary layers showing the paleohorizontal were found interbedded within the flows. For the Huambos formation the bedding was estimated from a structural study around the

site. As both volcanic formations are only slightly deformed, all the sampled sites are only gently inclined successions ($< 20°$). Significant declination errors due to tilt correction were thus avoided [13].

At each site a minimum of ten cores were drilled using standard drilling equipment and each core was independently oriented using both a magnetic and a sun compass.

3. Paleomagnetic results

3.1 General magnetic properties

The natural remanent magnetizations were measured on standard 22×25 mm cylindrical samples with either a spinner magnetometer or a LETI 3-axis cryogenic magnetometer. The highest magnetizations are those of the Huambos volcanic formations which range from 0.5 to 7 A/m (with the exception of site PE8723 which averages ~ 360 A/m, probably due to lightening). Lower magnetizations ranging from 0.01 to 0.6 A/m are observed for the intrusive formations.

Despite the wide differences in the lithologies of the formations studied, the magnetic mineralogy was found to be remarkably constant. Saturation of the Isothermal Remanent Magnetization (IRM) is attained below 0.25 T and the median AF destructive field of the SIRM is always lower than 3 mT, with the occasional presence of very small amounts of a higher coercivity material. During thermal demagnetization, the NRM decreases progressively and disappears at 580°C. Thermomagnetic curves were obtained from extracts using a horizontal Curie balance in a nitrogen atmosphere to avoid oxidation during the heating. Curves generated for all sampled lithologies show a monotonic decrease of Js up to 575° that is reversible upon cooling (Fig. 2). Thus, the results are all consistent with a magnetic mineralogy dominated by magnetite.

At each step of the thermal demagnetization, the low field magnetic susceptibility of every sample was systematically measured. No changes exceeding a factor of two were observed suggesting that no important mineralogical changes occurred during the thermal treatment in agreement with the reversible character of the thermomagnetic curves. The rock magnetic properties of the sampled formations are convenient for a paleomagnetic study.

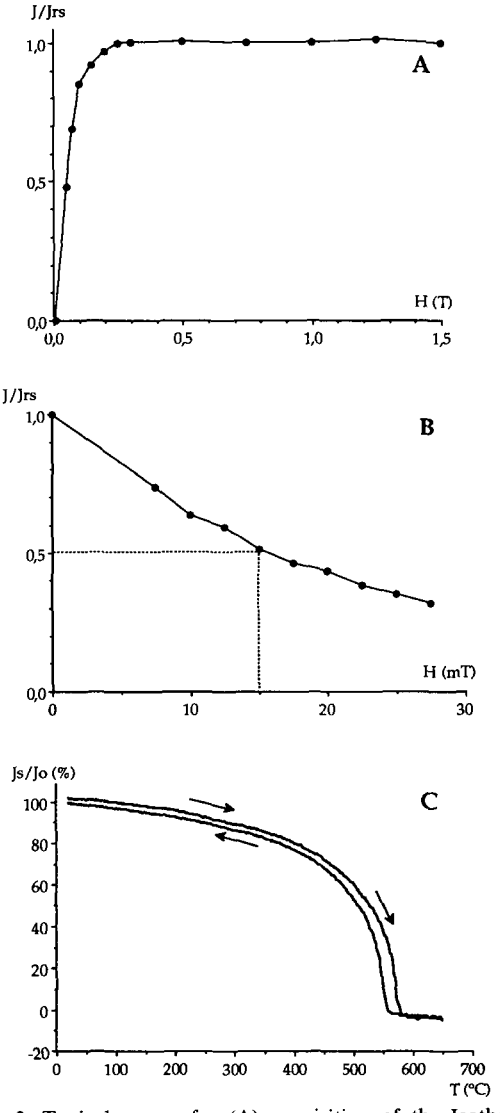


Fig. 2. Typical curves for (A) acquisition of the Isothermal Remanent Magnetization (I.R.M) and (B) thermal demagnetization of the Saturated I.R.M. The thermomagnetic curves obtained with a Curie balance (C) indicate a single Curie temperature at 575°C. Arrows indicate heating and cooling curves. All these results are consistent with a magnetic mineralogy dominated by magnetite.

3.2 Stable paleomagnetic directions

All samples were thermally demagnetized in ten to sixteen steps between room temperature and 530° to 580°C, only occasionally slightly higher.

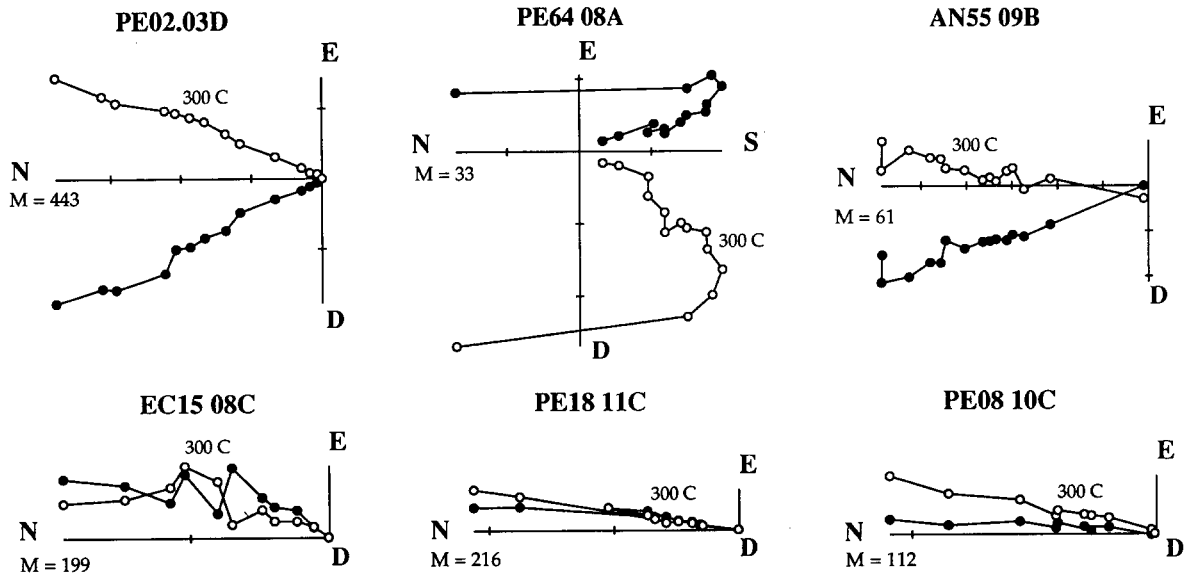


Fig. 3. Typical thermal demagnetization diagrams from the sampled formations. The 300°C step and the N.R.M intensity (10^{-3} A/m) are indicated on each diagram. Open circles and full circles correspond to vertical and horizontal component respectively.

The results were interpreted using orthogonal demagnetization diagrams some of which are shown on Fig. 3. At least the five to six last points of each diagram were used to determine the stable paleomagnetic directions either by hand or with a least square computer fit, the two methods yielding virtually identical results.

In general stable and consistent primary paleomagnetic directions were isolated after heating beyond 200°–300°C. Only occasionally, we rejected a very few samples from different sites with ambiguous demagnetization behaviour. However, three sites (EC87 13, PE87 03 and PE87 24) show erratic behaviour and no stable direction

TABLE 1

Reliable paleomagnetic results from the intrusive formations of the Lancones region

Sites	n	D (°)	I (°)	K	α_{95} (°)	Lithology
Lower intrusive unit						
AN86 73	12/12	38.0	−19.0	525	1.8	Las Lomas granodiorite
PE87 09	9/9	29.5	−25.7	76	5.3	Las Lomas granodiorite
PE87 11	10/10	30.5	−22.8	73	5.2	Las Lomas granodiorite
PE87 18	12/12	16.4	−9.8	208	2.8	Las Lomas granodiorite
EC87 15	7/9	22.0	−18.0	97	5.3	Macara intrusive unit
EC87 16	8/8	19.0	−17.0	26	9.7	Macara intrusive unit
Upper intrusive unit						
PE87 08	9/10	4.3	−9.0	378	2.4	Penablanca Monzogranite
PE87 15	8/10	2.0	−12.0	185	3.6	Montecillo granite
EC87 14	8/8	3.2	−7.7	34	8.4	Macara intrusive unit
EC87 01	8/8	185.0	38.7	68	6.0	Loja intrusive unit
Mean paleomagnetic direction						
	N = 10/10	D = 17	I = −18.3	K = 27	α_{95} = 8.4	
			$R = 24.7 \pm 11.6$		$F = 4.3 \pm 16.6$	

TABLE 2

Reliable paleomagnetic results from the volcanic and the intrusive formations of the Cajamarca region

Sites	n	Before bedding corr.		After bedding corr.		K	α_{95} (°)
		D (°)	I (°)	D (°)	I (°)		
Intrusives (no bedding correction)							
AN86 48	10/10	162.0	30.0	–	–	43	7.4
AN86 49	7/9	349.5	−22.5	–	–	41	9.6
AN86 54	9/10	329.0	−16.7	–	–	29	8.6
AN86 55	9/9	341.3	−11.2	–	–	50	6.5
AN86 56	6/7	342.4	−11.0	–	–	317	3.2
Mean direction:	$N = 5/5$	$D = 340.7$	$I = -18.4$	$K = 57.5$	$\alpha_{95} = 8.2$		
Llama volcanics							
PE87 25	8/8	323.0	−11.0	316.0.	−11.0	37	8.2
PE87 26 *	6/10	74.0	55.8	74.0	55.8	13	15.2
PE87 27	10/16	326.0	−43.0	330.0	−29.0	16	11.0
PE87 36	10/10	152.0	37.5	158.7	32.3	455	2.2
PE87 37	11/12	319.5	−45.0	318.3	−13.0	23	8.7
PE88 60 *	9/11	279.3	19.7	279.3	19.7	252	3.0
PE88 64	7/10	168.0	29.0	164.3	30.0	223	3.5
PE88 66	12/13	128.5	45.6	119.7	34.7	67	5.0
PE88 67	6/8	141.0	4.0	160.0	3.5	267	3.5
PE88 72	9/10	148.5	39.5	148.5	39.5	106	4.5
PE88 77 *	8/9	156.5	66.2	156.6	66.2	153	4.0
PE88 81	12/12	332.8	−29.0	344.7	−32.8	350	2.1
Mean direction (after BC):	$N = 9/12$	$D = 329$	$I = -25.8$	$K = 19.6$	$\alpha_{95} = 10.5$		
Huambos volcanics							
PE87 01	11/11	337.7	−14.3	333.6	−9.3	418	2.3
PE87 02	10/10	333.0	−19.7	329.0	−13.4	196	3.0
PE87 20	10/10	328.5	−23.0	326.0	−16.1	55	6.0
PE87 21	9/11	339.8	−17.5	336.0	−13.1	76	5.3
PE87 22	11/12	337.4	−16.0	337.4	−16.0	49	6.0
PE88 58	10/12	159.4	20.5	159.4	20.5	201	3.1
PE88 79 *	5/9	217.8	57.8	217.8	57.8	200	4.4
Mean direction (after BC):	$N = 6/7$	$D = 333.5$	$I = -14.8$	$K = 172.4$	$\alpha_{95} = 4.3$		

Mean regional direction taking into account the three formations after bedding correction:
$N = 20/24$ $D = 333.4$ $I = -20.6$ $K = 30$ $\alpha_{95} = 5.7$

$R = -19 \pm 9.7$ $F = 1.6 \pm 15$

could be determined from the demagnetization diagrams. Fisher's statistics was used to obtain the mean paleomagnetic directions and the statistical parameters at the different sites. In general, tightly grouped directions were observed. As an exception, four sites, two sampled near the Olmos massif, one from the Llama formation and one from the Huambos volcanics are characterized by an extremely high scatter, despite the almost rectilinear diagrams obtained from single samples. As mentioned before, the latter site has probably been struck by lightning. In total, only seven of the 41 sampling sites were not considered reliable and thus are not reported in Tables 1 and 2 nor in Fig. 4. Finally, directions from four sites (three from Llama, and one from Huambos) that diverged by more than two standard deviations from the mean regional direction were considered anomalous and consequently they were not included in the final statistics.

In the Lancones synclinorium the six sites belonging to the lower intrusive units are characterized by significantly easterly declinations. The four sites belonging to the younger, Si-rich unit, yield declinations that, although slightly easterly, do not significantly deviate from the geographical

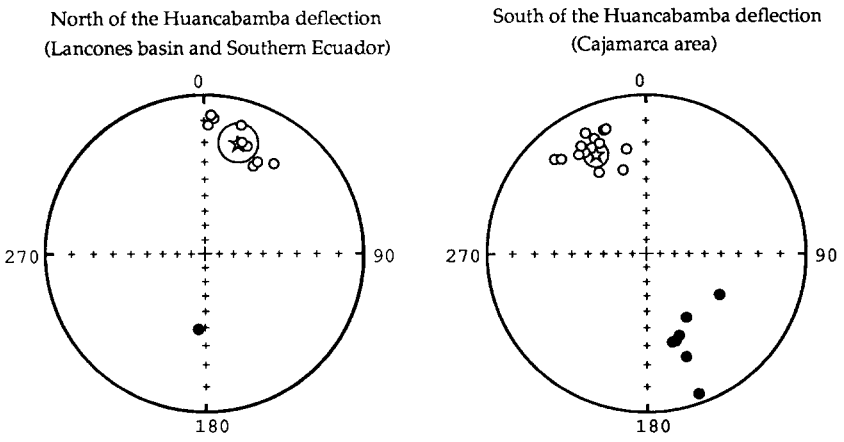


Fig. 4. Equal area stereographic projection of the mean direction of paleomagnetic vectors for the sampled formations on both sides of the Huancabamba deflection. Open circles correspond to upper-hemisphere and full circles to lower-hemisphere projections. Big circles correspond to the 95% confidence area about the mean (stars).

north–south axes. In both cases the inclinations are slightly steeper than expected on the basis of a centered dipole field. Undetected and different tilts at the different sites would cause larger scatter of the data, rather than coherent steepening of the inclinations and thus do not appear to be a reasonable explanation for a steeper inclination. Northward displacement or a general tilt affecting the entire region appear more probable. If this last hypothesis is correct, the steep inclinations can be accounted for by a small tilt ($< 10°$), thus, the obtained declinations would hardly be affected [13]. In any case, the steep inclinations give evidence that neither of the two units has been remagnetized. We have averaged the two groups in the final statistic ($D = 17$; $I = -18.3$; $K = 27$; $\alpha_{95} = 8.4$).

In the Cajamarca region the results from the twenty reliable sites sampled in three different lithological units are remarkably consistent. Most of the sites are characterized by normal polarity. However, when reverse magnetizations are observed, and this is the case for at least one site from each lithology, their directions are perfectly antiparallel to the normal ones. When the results from the Huambos, the Llama volcanics and the intrusive units are averaged together (inverting the reverse directions through the origin) they indicate a significant and clearly defined 27° westerly declination ($D = 333.4 \pm 6.1$).

4. Discussion

One of the difficulties encountered when trying to interpret the paleomagnetic data is the lack of a reliable Apparent Polar Wander Path (APWP) for stable South America (SOAM).

Recently the APWP has been greatly improved with the addition of four reliable poles for the late Cretaceous–early Tertiary (around 65 Ma) derived from volcanic and intrusive formations in Argentina and Patagonia [14,15]. When averaged, these poles yield a precise mean value defined by: $\lambda = 80.7S$; $\phi = 355.8E$; $A_{95} = 4.7$. However, for the entire Cenozoic the only available poles are still those given by Irving and Irving [16] for 15, 8 and 4 Ma, substantially younger than the ages of the formations considered in this paper. We note however that the 15 Ma pole of Irving and Irving ($\lambda = 83S$; $\phi = 345E$; $A_{95} = 10.5$) is rather close to the mean 65 Ma pole given above, the main difference being the better precision of the latter. The movement of SOAM thus appears to have been rather limited between the two epochs. For the interpretation of the Eocene–Early Oligocene data, one could then reasonably use either one of the two poles. We have however preferred to calculate an intermediate 40 Ma pole averaging out the new mean 65 Ma pole and the 15 Ma pole used by Irving and Irving. We have then referred all our results to this calculated pole ($\lambda = 81.9S$; $\phi =$

350.4E; $A_{95} = 7.5$). We must stress that the interpretation of the results would be unchanged if the mean late Cretaceous–early Tertiary pole or the 15 Ma pole of Irving and Irving had been used instead. The rotation (R) and flattening (F) parameters calculated with this pole using the equations given by Beck [17] and Demarest [18] are given in Tables 1 and 2 for the sites north and south of Huancabamba respectively. The large error values associated to the two parameters and particularly to F are mainly due to the large value of A_{95}.

South of the Huancabamba deflection, the very consistent paleomagnetic results obtained in the Cajamarca area from the Llama and the Huambos volcanic formations and from the coastal intrusive stock clearly demonstrate the occurrence of a 19° ± 9.7° post-early Oligocene counterclockwise rotation affecting the coastal area of northern Peru. The sense of this rotation agrees with previously published paleomagnetic results obtained in the same area from Cretaceous formations [7] and also with those obtained from Mesozoic formations farther east in the Bagua Grande area [3] which indicate ~ 30° anticlockwise rotations. Our results thus show that a substantial fraction of the rotation recorded by Mesozoic formations is much more recent than previously believed.

North of the Huancabamba deflection, the results from the Lancones Basin in northern Peru and southern Ecuador consistently show a clockwise rotation, the amplitude of which is 25° ± 11.6° when the mean paleomagnetic direction is referred to the 40 Ma reference pole.

The results reported here thus document an unpredicted pattern of post-Paleocene clockwise and post-early Oligocene counterclockwise rotations north and south of the Huancabamba deflection, respectively. So far, only one study suggesting a ~ 15° counterclockwise rotation has been reported from Neogene post-tectonic rocks in the Central Andes of Peru, the Ocros dyke swarm near Ayacucho [19]. However, Beck [5] has shown that these results do not meet simple reliability criteria so that they must be considered with some caution. Thus, to our best knowledge, our results are the first to unambiguously document post-early Oligocene rotations in the Central Andes.

This pattern of clockwise and counterclockwise rotations could be due either to block rotation in a zone of distributed shear, or to oroclinal bending of the Cordillera. For the regions studied here, distributed dextral and sinistral shear respectively north and south of the Huancabamba deflection could arise as a result of the different angular relation between the trend of the margin and the direction of the convergence, which has been closely E–W in the last 28 Ma [20,21]. In such a tectonic regime, which would lead in each region to rotations of the observed sense, one would expect the shear to be associated with strike-slip faulting, which should be visible in the field in both studied regions. In the Cajamarca region, strike-slip faults have been documented at the borders of the Mesozoic "intra-chain" barriers and along the fold axis of the Mesozoic formations [22–24]. However, these faults do not have large amplitude offsets nor are they widespread and thus cannot account for the large measured rotations.

In the Lancones basin post-Cretaceous active faulting has been documented, but has been considered so far as purely extensional [8], although at the edges of the Amotape range, the steeply dipping folds axes may be related to strike-slip faulting. Clear evidence for a dextral strike-slip regime is, however, present immediately north of the sampled region in southern Ecuador. The main Dolores-Guayaquil megashear has been documented by many geological and geophysical studies [25,26] and smaller faults with Oligocene and Miocene dextral strike-slip movements have been recognized just north of the Lancones Basin [27] suggesting that a large zone of distributed shear might exist to the east of the megashear.

The tectonic regimes north and south of the Huancabamba deflection are thus quite different and there is non a-priori reason to think that the rotational patterns prevailing in these two regions arise from the same mechanisms. In our opinion the clockwise rotation north of the bend may reasonably be related to the distributed shear regime documented by geological studies. South of the bend, on the contrary, the absence of any clear evidence of large scale left lateral strike-slip faulting over the region affected by the rotation, suggests that the results reflect a rotation of the Peruvian margin, rather than a distributed shear.

Recently, Isacks has proposed a model relating the uplift of the Central Andean Plateau to the

bending of the Bolivian orocline [28]. In this model, along-strike variations in the amount of late Cenozoic shortening result from corresponding variations of the width of a weakened zone in the overriding plate. Maximum shortening exists at 20°S latitude and decreases both northward and southward leading to an enhancement of the concave shape of the Arica elbow and to counterclockwise and clockwise rotations north and south of it. When combined with the available paleomagnetic results from the Central Andes, the model predicts a maximum shortening significantly larger (425 km) than the one documented by the geological studies (210–250 km [29]). Isacks argues that the paleomagnetic data are from Mesozoic formations and thus integrate the deformation over the entire Cenozoic, while the model only considers late Cenozoic evolution. The results reported here, which show that about 60% of the total rotation has occurred during the Cenozoic allow a more realistic evaluation. When the value of the post-early Oligocene rotation is used, the model predicts a maximum shortening of about 350 km more in agreement with the geological estimate, especially considering that this latter gives a lower bound for the shortening (note that were the Peruvian margin rigid, the observed 19° ± 9.7° counterclockwise rotation would imply a difference in shortening of 530 ± 270 km between northern and southern Peru).

Although Isacks' model appears to yield rather satisfactory results, we must stress that the hypothesis of a relationship between the rotations reported here and the uplift of the Andes depends on the extent to which our results, obtained in the region of the Cajamarca deflection, are significant for the entire Peruvian margin. It depends also on the precise timing of the rotation which is at present not better constrained than post-early Oligocene. Indeed the Andean uplift is entirely Neogene, and according to Isacks, only extension without any shortening has occurred from early Oligocene to early or middle Miocene. So far, only two paleomagnetic studies have been conducted in recent formations in the Central Andes. In one case [30], no rotation was detected from Miocene and Pliocene formations in southern Peru, in agreement with the predictions of Isacks' model. In the second case, preliminary results obtained from lacustrine basins in Bolivia have reported that no significant rotation has affected this area during the last 25 Ma [31]. Those results would appear inconsistent with Isacks' model. However, the investigated region is close to the Arica elbow where the pattern of counterclockwise rotation in the north changes into clockwise rotations in the south. The amount of rotation in this region depends upon the detailed trajectories of material in the deforming forearc. A thorough investigation of the structural setting of the sampled sites is thus needed to assess the bearing of these preliminary results upon the movement of Central Andes in Peru.

Thus our results need to be extended southward, with particular attention to Neogene formations in the Central Andes, before the relation between the rotations, the maximum shortening, and the uplift can be considered as definitively established.

5. Conclusions

The results obtained here from post-Paleocene and early Oligocene formations in southern Ecuador and northern Peru, although certainly incomplete, yield convincing evidence for a pattern of clockwise and counterclockwise rotations respectively north and south of the Huancabamba deflection. While in the north, these rotations seem to be related to block rotation in a distributed dextral shear, in the south, the rotational pattern results more realistically from a rotation of the entire Peruvian margin. This rotation is most probably related to the uplift of the Central Andean Plateau, in agreement with recent published model relating this uplift to the bending of the Bolivian orocline. When combined with this model, the data obtained here suggest that the amount of shortening in the Central Andes is greater than documented by geological studies, but a more detailed evaluation will only be obtained through additional paleomagnetic investigations of Cenozoic formations in Central Peru.

Acknowledgements

This work is part of a cooperation between the CFR and the Institut Français d'Études Andines (IFEA). We wish to thank Yves Saint-Geours, Director of the IFEA, for his help in the organisa-

tion of the field work. F. Mégard and T. Mourier participated to the sampling and to many stimulating discussions. We thank P. Molnar for his critical reading of the manuscript. We also acknowledge many discussions with P. Roperch and O. Bellier. A. Roberts and D. Schneider kindly improved the quality of the manuscript. The financial support was given by the CEA, the CNRS, the IFEA and by the INSU-CNRS program DBT–Dynamique Globale. This is the DBT–Dynamique Globale contribution 128 and the CFR contribution 1084.

References

1 H.C. Palmer, A. Hayatsu and W.S. McDonald, The Middle Jurassic Camaraca Formation, Arica, Chile: Paleomagnetism, K-Ar dating and tectonic implications, Geophys. J. 62, p. 155, 1980.
2 K. Heki, Y. Hamano, H. Konoshita, A. Taira and M. Kono, Paleomagnetic study of Cretaceous rocks of Peru, South America: evidence for rotation of the Andes, Tectonophysics 108, p. 267, 1984.
3 M. Kono, K. Heki and Y. Hamano, Paleomagnetic study of the Central Andes: counterclockwise rotation of the Peruvian Block, J. Geodyn. 2, p. 193, 1985.
4 S.R. May and R.F. Butler, Paleomagnetism of the Puente Piedra Formation, central Peru, Earth Planet. Sci. Lett. 72, p. 205, 1985.
5 M.E. Beck Jr, Analysis of late Jurassic–Recent paleomagnetic data from active plate margins of South America, J. S. Am. Earth Sci., 1 (1) p. 39, 1988.
6 T. Mourier, C. Laj, F. Mégard, P. Roperch, P. Mitouard and A. Farfan Medrano, An accreted continental terrane in northwestern Peru, Earth Planet. Sci. Lett. 88, p. 182, 1988.
7 C. Laj, P. Mitouard, P. Roperch, C. Kissel, T. Mourier and F. Mégard, Paleomagnetic rotations in the coastal areas of Ecuador and Northern Peru, in: C. Kissel and C. Laj, eds., Paleomagnetic Rotations and Continental Deformation, Kluwer, Amsterdam, p. 489, 1989.
8 G. Beaufils, Recherche de concentration metalliques sulfurées et cartographie géologique dans la région de Tambo Grande, BRGM 007/78, Lima, 1978 (unpubl.).
9 T. Mourier, F. Mégard, A. Pardo and L. Reyes, L'évolution mésozoïque des Andes de Huancabamba (3°–8°S) et l'hypothèse de l'accrétion du bloc Amotape–Tahuin, Bull. Soc. Geol. Fr. 8, p. 69, 1988.
10 R. Luis Reyes and V. Julio Caldas, Geologia de los cuadrangulos de Las Playas, La Tina, Las Lomas, Ayabaca, San Antonio, Chulucanas, Morropon, Huancabamba, Olmos y Pomahuaca., Boll. Inst. Geol. Min y Metall. 39, A, 1987.
11 C.C. Noble, E.H. McKee, T. Mourier and F. Mégard, Cenozoical stratigraphy, magmatic activity, compressive deformation and uplift in Northern Peru., Geol. Soc. Am. Bull., 1990 (in press).
12 J.W. Stewart, J.F. Evernden, N.J. Snelling, Age and determination from Andean Peru: a reconnaissance survey, Geol. Soc. Am. Bull. 88, p. 1107, 1974.
13 W.D. McDonald, Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res., 85, p. 3659, 1980.
14 C.R. Montes-Lauar and I.S. Pacca, Palaeomagnetism of alkaline intrusions in the South American platform and their possible relation with hot spot tracks, abstract 6th IAGA Scientific Assembly, Exeter, 1989.
15 R.F. Butler, F. Hervé, F. Munizaga, R.F. Burmester, M.E. Beck and E.SD. Oviedo, Late-Cretaceous–Early Tertiary Paleomagnetic Pole from the Patagonian Basalts, southern Chile and Argentina. Abstract AGU Fall Meeting, EOS, 1989.
16 E. Irving and G.A. Irving, Apparent Polar Wander Paths Carboniferous through Cenozoic and the Assembly of Gondwana., Geophys. Surv. 5, p. 141, 1982.
17 M.E. Jr. Beck, Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, J. Geophys. Res., 85, p. 7115, 1980.
18 H.H. Demarest Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, J. Geophys. Res. 88, p. 4321, 1983.
19 K. Heki, Y. Hamano, M. Kono and T. Ui, Paleomagnetism of Neogene dike swarm, the Peruvian Andes: implication for the Bolivian Orocline., Geophys. J. R. Astron. Soc. 80, p. 527, 1985.
20 R.H. Pilger Jr., Kinematics of South American subduction zone from global plate reconstructions, in: Geodynamics of the Eastern Pacific Region. Caribbean and Scotia arcs, R. Cabre ed., Geodyn. Ser., Am. Geophys. Union, 113 pp, 1983.
21 F Pardo Casas and P. Molnar, Relative motion of the Nazca (Farallon) and South America Plates since late Cretaceous time, Tectonics 6, p. 233, 1987.
22 D. Janjou, Données géologiques pour un modèle d'évolution des Andes nord-péruviennes, Mem. Sci. Terre, Univ. P. et M. Curie, Paris, 81–35, 170 pp., 1981.
23 T. Mourier, La transition entre Andes Marginales et Andes Cordilleraines à ophiolites: évolution sédimentaire, magmatique et structurale du relais de Huancabamba (3° à 8°S: Nord Pérou–Sud Equateur). Thèse d'Etat, Univ. Paris-XI Orsay, France, 301 pp, 1988.
24 O. Bellier, Tectonique en extension et changement d'état de contraintes Cénozoïque en domaine intracontinental: Exemple des bassin intra-cordillerains des Hautes Andes (Nord-Pérou) et du graben de la Wei He (Chine du nord). Thèse de Doctorat, Univ. Paris-XI Orsay., France, 300 pp, 1989.
25 F. Mégard, Cordilleran Andes and Marginal Andes: a review of Andean geology north of the Arica elbow (18°S), in: Circum Pacific Orogenic Belts and Evolution of the Pacific Ocean Basin, J.W.H. Monger and J. Francheteau, eds., Am. Geophys. Union, Geodyn. Ser. 18, p. 71, 1987.
26 T. Feininger and C.R. Bristow, Cretaceous and Paleogene geologic history of coastal Ecuador, Geol. Rundsch. 69, p. 849, 1980.
27 C. Noblet, A. Lavenu and F. Schneider, Étude géodynamique d'un bassin intramontagneux tertiaire sur décrochements dans les Andes du sud de l'Equateur: l'exemple du bassin de Cuenca. Rapp. ORSTOM, Paris, 50 pp, 1988.

28 B. Isacks, Uplift of the Central Andean Plateau and Bending of the Bolivian Orocline, J. Geophys. Res. 93, 3211, 1988.
29 B.M. Sheffels, Structural constraints on crustal shortening in the Bolivian Andes, Ph.D. Thesis, M.I.T., Cambridge, Mass., 165 pp., 1988 (unpubl.).
30 H. Tsunakawa, H. Tanaka, K. Amano and M. Kono, Paleomagnetic study of Late Miocene and Early Pliocene rocks from southern Peru, Central Andes, J. Geomag. Geoelectr. 39, p. 477, 1987.
31 B.J. McFadden, Paleomagnetism and chronology of Late Cenozoic Andean basins and comments on the Bolivian orocline hypothesis, Abstract AGU Spring Meet., 1989.